Madhav Chopra

Senior Associate | Full-Stack Developer & Python Specialist

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Summary

Experienced Python Developer with 5 years of expertise in full-stack development, system migrations, and data engineering. Skilled in leading cross-functional teams to streamline data extraction, sanitization, and dynamic web application development using modern Python frameworks (FastAPI, BeautifulSoup, Pandas).

Skills

Languages: Python (3.12+), JavaScript, SQL, Bash/Shell Scripting, HTML5/CSS3

Frameworks & Web: FastAPI, Django, ReactJS, Pydantic, Selenium, Playwright, Beautiful Soup

Data & ML: Polars, Pandas, NumPy, Scikit-Learn, NLTK, Matplotlib, Seaborn, Jupyter

Cloud & DevOps: Docker, Docker Compose, Ansible (IaC), Azure DevOps, Git, Nginx, Linux Administration

Virtualization: Proxmox VE, LXC/LXD Containers, Portainer

Core Engineering: Multithreading, Asyncio, REST APIs, System Design, Agile

Experience

Senior Associate, PwC India – Gurugram, Haryana Apr 2026 – present

Backend engineer specializing in the FastAPI ecosystem, delivering end-to-end backend solutions from initial design through deployment for projects including APMT and SEPS.

- Architected and delivered production-grade REST APIs using **FastAPI** for Adani's **APMT** and **SEPS** projects, providing complete backend support.
- Designed scalable backend architectures leveraging **Pydantic** for data validation, **asyncio** for high-throughput request handling.
- Collaborated closely with cross-functional teams to translate business requirements into technical solutions, maintaining high code quality and adherence to **Agile** best practices.

System Engineer, Tata Consultancy Services – Gurugram, Haryana July 2021 – Apr 2026

Hand-picked for the Rollout Team to lead website migrations for enterprise clients. Built Python-based ETL pipelines for automated data extraction and sanitization, while delivering full-stack features with **ReactJS** and **Python**.

- Built migration automation suite using **Polars**, **Pandas**, and **BeautifulSoup**, cutting delivery timelines by **50%** and doubling data throughput.
- Created an **SEO Checker** tool to validate migration accuracy with automated Excel reports, reducing manual QA by **40%**.
- Optimized **Core Web Vitals** and achieved **WCAG accessibility compliance** for production websites.
- Recognized as **"Presentation Hero"** for clear technical communication with international clients.

Projects

SEO & Link Compliance Auditor

Full-featured desktop application for validating website link integrity and SEO compliance, featuring both CLI and GUI interfaces with automated reporting.

- Built a cross-platform desktop SEO auditor using **Playwright** for async scraping, with a **link caching system** and URL normalization to cut redundant requests.
- Implemented dual-interface design (CLI for CI/CD, Tkinter GUI) with **Pytest** coverage.
- Automated Excel report generation via **Pandas** and **OpenPyxl** with conditional formatting.

Automated Infrastructure

Built and maintained a self-hosted lab for service testing and deployment.

- Automated server configuration across Linux LXC containers using **Ansible** playbooks.
- Orchestrated services via **Docker Compose** and **Portainer** for high availability.
- Hardened security with **UFW** firewalls, SSH key auth, and Nginx reverse proxy.

Education

Punjab Technical University, Bachelor of Technology in Computer Science and Engineering — May 2021

Certifications

Generative AI & LLMs: Azure OpenAI & GenAI Models, Prompt Engineering, Building Custom GPTs

Data Science & ML: OpenEDG Python Professional, Scikit-Learn, Pandas to Polars, Matlab ML

Cloud & DevOps: Azure DevOps Fundamentals (Udemy), DevOps Foundations, Git Essential Training

Programming & Tools: Linux Shell Scripting, JavaScript (ES6+), Vim, Machine Learning (Stanford/Coursera)

Awards

Beyond Performance Award — Sep 2024 & Jan 2023

Service & Commitment Award — July 2024

Star of the Month — Apr 2022